Experiment 7: Android Calculator Application

AIM:

Todevelop an Android application that performs basic arithmetic operations such as addition, subtraction, multiplication, and division using Android UI components.

PROCEDURE:

1. Open Android Studio and create a new project.
2. Choose Empty Views Activity template.
3. Design UI using EditText, Buttons, and TextView in XML.
4. Link UI components in MainActivity.java.
5. Retrieve user inputs and convert to numbers.
6. Set OnClickListeners for operation buttons.
7. Perform arithmetic operations.
8. Display result in TextView.
9. Handle invalid inputs and division by zero.
10. Run the application using emulator.

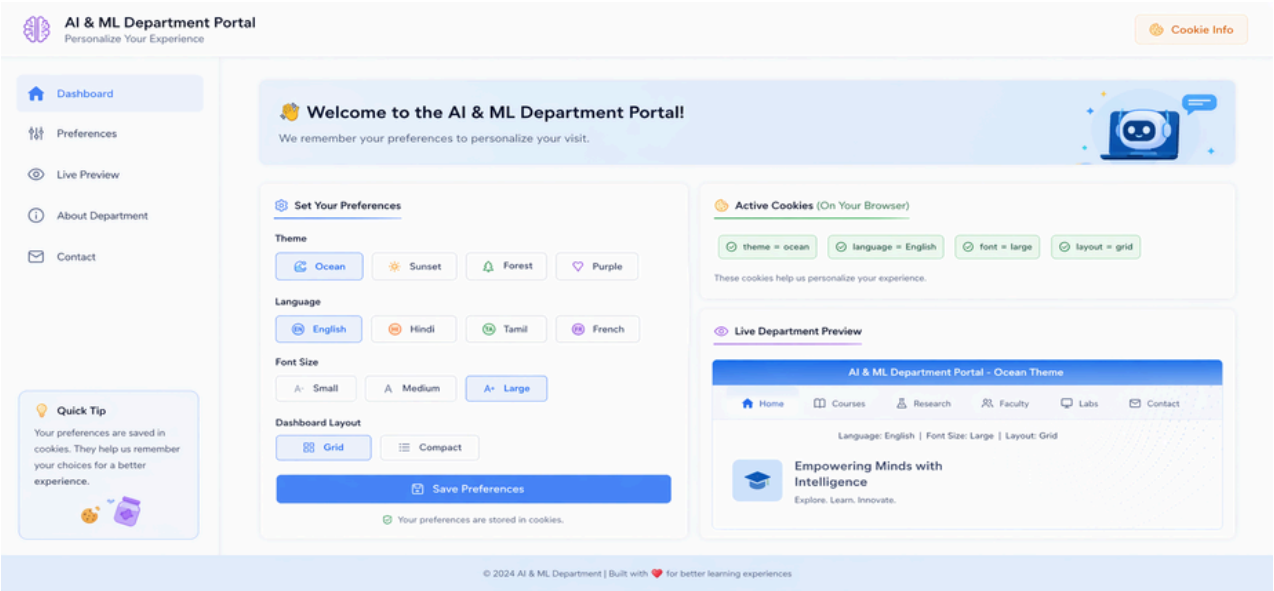
CODE:

XMLLayout:

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class PreferencesServlet extends HttpServlet {

   protected void doGet(HttpServletRequest request, HttpServletResponse response)
         throws ServletException, IOException {

      response.setContentType("text/html");
      PrintWriter out = response.getWriter();

      String theme = "Not Set";
      String language = "Not Set";
      String font = "Not Set";

      Cookie[] cookies = request.getCookies();

      if (cookies != null) {
         for (Cookie c : cookies) {
            if (c.getName().equals("theme")) theme = c.getValue();
            if (c.getName().equals("language")) language = c.getValue();
            if (c.getName().equals("font")) font = c.getValue();
         }
      }

      out.println("<h4>Theme: " + theme + "</h4>");
      out.println("<h4>Language: " + language + "</h4>");
      out.println("<h4>Font: " + font + "</h4>");
   }

   protected void doPost(HttpServletRequest request, HttpServletResponse response)
         throws ServletException, IOException {


}
```

```
String theme = request.getParameter("theme");
 String language = request.getParameter("language");
 String font = request.getParameter("font");

 Cookie c1 = new Cookie("theme", theme);
 Cookie c2 = new Cookie("language", language);
 Cookie c3 = new Cookie("font", font);

 c1.setMaxAge(86400);
 c2.setMaxAge(86400);
 c3.setMaxAge(86400);

 response.addCookie(c1);
 response.addCookie(c2);
 response.addCookie(c3);

 response.sendRedirect("index.html");
 }



}
```

OUTPUT:



RESULT:

An Android calculator application is successfully developed to perform basic arithmetic operations with proper input validation.